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AI-backed OCR in Healthcare

Daniela Gifu[a,b,*]

[a]Faculty of Computer Science, "Alexandru Ioan Cuza" University, General Berthelot, 16, 700483, Iasi, Romania
[b]Institute of Computer Science, Romanian Academy - Iasi branch, Bulevardul Carol I, 8, 700505, Romania

Abstract

As revolutionary technology, scanning technology - like OCR - knows an attracting increasing interest in the medical system. In fact, scanning technology, driven by big data and machine learning, helps to drive successful change processes in healthcare. Nowadays, the OCR systems based on the promise of artificial intelligence can contribute to a greater understanding of entire processes by automating the medical transcription and in general mining information (largely handwritten) clinical notes. The project aims to create an OCR application for offline character recognition using deep learning (a combination of RNN and CNN) that can be used in the medical sector. Concerning problems of medical care, large data such as dispersed dataset, weak consistency, low electronic degree, and low visualization degree, a set of innovative image recognition, dataset, arrangement, and storage projects were put forward. The proposed pilot method achieves results comparable to current SoTA.

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Keywords: clinical notes; OCR; classification; deep learning; accuracy;

1. Introduction

Given the ubiquity of handwritten documents that appear in daily activities in different domains, like the medical field, Optical Character Recognition (OCR) has invaluable practical worth [1]. Since the development of artificial intelligence (AI), which responds to people's need to simplify their interactions or transactions [2; 3], the interest in

* Corresponding author. Tel.: +40 232 201771
E-mail address: daniela.gifu@info.uaic.ro

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building character recognition architectures it is constantly on the rise [4]. Today, Tesseract and Google Vision are considered the most accurate OCR engines [5], but this paper is by no means a comparative analysis.

There is a variety of ways of developing classifying systems, as in case the OCR classification [6], each with its own set of goals, but all aiming to produce models that are as similar to humans as possible [7]. In this context, Natural Language Processing (NLP) [8] is best applicable for solving classification of the medical records [9; 10; 11], which can improve the doctor-patient relationship [12].

Basically, automatic text recognition in images or scanned documents by OCR aims to improve the chances of successful recognition [13] by utilizing image preprocessing techniques (e.g., image operations like resizing, cropping, rotating, color conversion, etc.), which converts image data into a form that allows AI algorithms to solve it [14; 15].

In this paper, we propose a pilot OCR application for offline handwriting recognition using Deep Learning (DL) - a combination of Recurrent Neural Network (RNN) and Convolutional Neural Network (CNN) - useful for speeding up the process of filing and accessing medical records, claims, or about any other possible medical form. It is well known that the offline handwriting recognition means character recognition and word recognition. Once these meaningful units of language are written down, they won't have information to the strokes/directions involved during writing. We refer to a possible addition of some background noise from the source (e.g., paper). Furthermore, the tasks of this application provide a solution for recognition predicated on touch input, recognition from live camera frames or a picture file, learning incipient characters and learning interactively predicated on the utilizer's feedback, many of them also targeted by other researchers [16].

The main research questions of this paper intend to answer: *Why can nobody understand the handwriting of a doctor? How AI is enhancing medical records scanning?*

Overall, the contributions of this work are twofold: (i) A pilot system based on a combination of two DL models useful to identify and classify digital text into editable one in the Romanian medical field. The correlative studies, comparing to other well-known OCRs, showed a statistical significance that contribute to identify characters in the medical field and classify them to be editable and easy to recognize with high accuracy. (ii) This model is statistically comparable to competitive SoTA.

2. A Brief OCR History in Healthcare

Since the 1990s, Handwriting Text Recognition (HTR) remains a challenging problem [17] for computer scientists and as work in progress, it is not yet fully automated nor flawless. To recognize an individual's handwriting is still a complicated process.

This section presents briefly several handwriting recognition systems, which proves their efficiency in the medical sector. Modern OCR engines, developed and maintained by Google, like Tesseract [18; 19] and Google Vision [20], have a long start over their competitors. These tools provide support for many languages (over 100). The overall percentage of recognition of handwriting characters to both tools is ≤ 90%.

In the medical field, a handwriting recognition system is vital, as hundreds of millions of medical claims are processed each year, resulting in a significant amount of paperwork. Basically, manual processing, full of medical terminology, that most people cannot understand, becomes impossible to complete. Prior approaches have fixated on individual aspects of the task complexity. However, NLP components that require dictionaries and other comprehensive corpora are problematic in some contexts. For instance, a separate neural language model is often trained to match the source language of the documents being considered. It takes the output of a vision-predicated transcription network and makes rectifications predicated on a learned probability distribution of character or word-level sequences [16]. Bellow, the State-of-The-Art (SoTA) architectures are briefly discussed.

2.1. Character recognition with CNN

Two methods are very common and make a huge impact in document transcription. Both of them are based on DL or probabilistic generative models. A feed-forward CNN that determines the lexical attributes of a word for transcription purposes [21] is introduced. Similar approaches, in general, for font classification [22], were used for offline Persian character recognition [23] and compared Siamese and triplet networks and showed performance improvement when combined with a CNN for handwritten Urdu character recognition [24].

CNN proved its strength in document recognition tasks [25]. It was trained using samples of a standard 50-class Bangla basic character database and features have been extracted for 5 different 10-class numeral recognition problems of English, Devanagari, Bangla, Telugu and Oriya each of which is an official Indian script [26].

2.2. Character recognition with CRNN

For historical documents, knowing the degraded quality of the paper, to train a high-quality OCR model is a challenging work. The Convolutional Recurrent Neural Network (CRNN) [27] generates better results than other models. For instance, it consists of five convolutional layers and three bidirectional LSTM layers and uses a learning rate of 15 epochs.

A few studies examined recurrent approaches for sequence processing by using Bidirectional Long Short-Term Memory (Bi-LSTM) in different input and architectural configurations. The final scope is to combine these insights by replacing regular neural network units with memory cells [28; 29]. Each memory cell contains a central linear recurrent unit with an untrainable weight of one, and a non-recurrent unit with a non-linear activation function processes the central unit's incoming information. Here, RNNs with LSTM units are particularly good for handwriting recognition. In fact, the SoTA systems, for many public databases, include an RNN component. For instance, it was trained a bidirectional LSTM-RNN (BLSTM-RNN) on sequences of feature vectors [30].

Moreover, CRNN for HTR has been constantly breaking SoTA results and been deployed in industrial application [31]. It also used 3x3 pixels and a number of 16 filters at the n-th convolution layer. To reduce the overfitting, it was applied in its system Dropout at the input for the convolutional blocks [32]. On the recurrent blocks, Puigcerver system is formed by bidirectional LSTM layers and uses a number of 5 recurrent blocks and a minimum of 20 epochs of training and maximum 80.

From the underlined ones, CRNN is becoming very useful for text detection, segmentation, and recognition using mostly images without detection or segmentation annotations. For instance, the Start, Follow, Read (SFR) model is composed of a Region Proposal Network to find the start position of text lines, a novel line follower network that incrementally follows and preprocesses lines of (perhaps curved) text into dewarped images suitable for recognition by a CNN-LSTM network [33; 34]. There are multiple DL approaches for OCR [35; 36] like: multiple sequence tagging [37; 38], character-level detection [39] or a combination between word-level and character-level features [40], etc.

A comparison between CNN and CRNNs on recognition of handwritten characters from the medical field proves to be more than satisfactory, being also the path chosen for this study.

3. Materials and Methods

Due to the need for medical process, many OCRs have been developed to serve the automated transcription of handwritten documents in healthcare (Fig. 1).

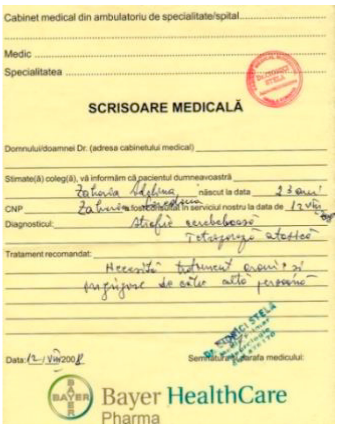
Cabinet medical din ambulatoriu de specialitate/spital
Medic
Specialitatea
SCRISOARE MEDICALĂ
Domnului/doamnei Dr. (adresa cabinetului medical)
Stimate(ă) coleg(ă), vă informăm că pacientul dumneavoastră
născut la data
CNP
Diagnosticul
Tratament recomandat
Data:
Semnătura și parafa medicului:

Fig. 1. Sample letter of medical necessity.

The novelty element of this paper is the construction of a medical corpus for the Romanian language. It contains 7,700 handwritten samples of 214 words. These medical records are obtained from 17 physicians. One-thirds of 214 words are randomly selected from the complete test data. Thus, the train data has 5133 and the test data has 2567 handwritten samples. The data augmentation methods are applied only to the train data.

For text classification task, we used the Kaggle library, because it contains images with 26 letters (A through Z) plus other signs for *space*, *delete* and *nothing*.

3.1. Materials

Kaggle library contains all 26 letters (or characters) in separate folders. Each folder has over 1000 images per letter and a total of 372451 images. In this library, the number of images is not equal per each class (varies between the number of images corresponding to the letter O = 57825 and the letter Y = 1085) (Table 1).

This might influence the problem of deliverability inaccuracy.

Table 1. The 26 letters of American Sign Languages (ASL) – number images.

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
Number images	14780	8976	23408	10133	11439	11620	5761	7217	1097	8943	5063	11586	12336
Letter	N	**O**	P	Q	R	S	T	U	V	W	X	**Y**	Z
Number images	19010	**57825**	19341	5812	48419	11580	22495	29008	4812	10784	6272	**1085**	6079

The letters from folders have been read in 26 different classes with an ImageDataGenerator tool from Keras that rescale all the images to the same dimensions of 1/255. The rescaling is mandatory because the original images consist of RGB pixel values on a scale of 0 to 255. Such values would be too high for our model to process (given a typical learning rate). We target values between [0,1] instead of by scaling with a 1/255 (Fig. 2).

Fig. 2. Number of images for each letter followed by a letter example.

After the reading in all 26 classes, we will have the images of the same dimensions for each letter that are ready to be trained. The same reading of classes was used for both models.

3.2. Architecture

Figure 3 presents the design flow (highlighting the steps covered in this paper), described briefly below.

A handwriting recognition system consists of several steps: handwritten dataset, data preprocessing, increasing data samples using data augmentation, and building a deep learning model for predicting doctors' handwriting.

Before recognizing text in scanned medical records, it is necessary to perform several document analysis operations like thresholding; noise removal; line segmentation; word and character or letter segmentation.

The basic problem is to assign the digitized character to its symbolic class. Here, the letter is labeled in 26 classes and the network is trained. After the training of the corpus, the testing images are used. The workflow starts reading the letters and creating the dataset training. Then, we train the network according to the approach using CNN or CRNN with dataset collected previously. A label function is applied to the output of the network.

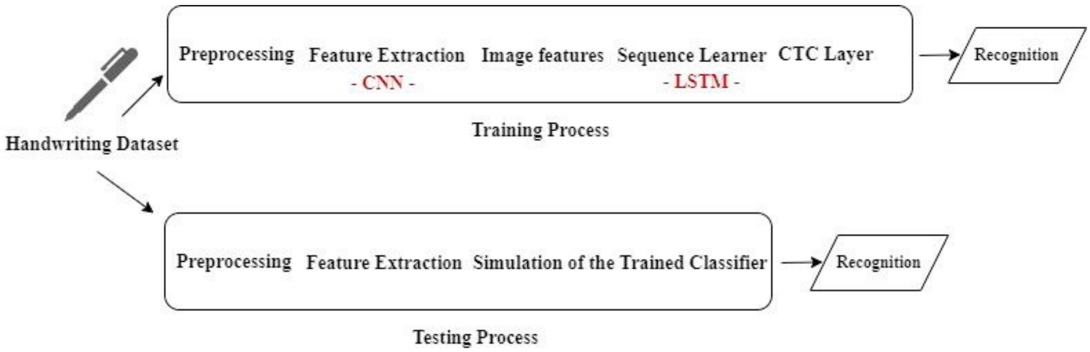

Fig. 3. System architecture.

The output contains a vector of 26 positions, and the position with the highest number is supposed to be the predicted letter like in the image below (Fig. 4).

```
[[5.4186158e-04 4.8729260e-03 8.2049137e-01 2.2869220e-02 4.1104611e-03
  1.2841793e-03 3.4477853e-03 2.2398289e-03 1.8063707e-03 1.3043307e-02
  2.4016569e-03 7.1904291e-03 7.0504355e-03 1.1054720e-02 1.8850293e-02
  2.5069623e-03 1.1885368e-03 2.1307908e-03 2.0422105e-02 2.4418586e-03
  3.5097308e-02 7.7576581e-03 2.0401967e-03 1.4014813e-03 2.9563685e-03
  8.0185465e-04]]
```

Fig. 4. Output of the neural network

Given an image from the testing resource, the output is predicted based on the network and labeled.

3.3. Used Technologies

In this study, we set up Jupyter Notebook, which provides smart coding support for Python. Then, Keras was installed. Basically, users are allowed to run code on powerful CPUs and GPUs, while using a Python 3.7 engine.

3.4. Deep Learning Approaches

In contrast to human brains, which can recognize and memorize characters such as letters or digits, computers treat them as binary graphics. As a result, algorithms are required to identify and recognize each character.

To achieve high recognition performance of handwritten special character in the medical field, we apply offline methods of recognition at every character, based on a sequence-to-sequence approach with CNN and CRNN networks.

Below, each neural network approach is described separately.

- **Convolutional Neural Network (CNN) –** is a DL algorithm which can take an input image and assign importance to various aspects of it. Regarding the architecture used in the current implementation, the CNN used a convolutional layer followed by a pooling layer. A second convolutional layer is followed again by a polling layer. Both convolutional layers have activation type Rectified Linear Unit (ReLU). Finally, the output is filtered by a Dense layer. In order to accelerate the gradient descent algorithm, Adam optimizer [41] was chosen. The number of training epochs was 18. The core component of a CNN is its convolutional layer. The layer's parameters are composed of learnable filters that extend through the full depth of an input. These components are then computed and activated by the network. The full output volume of the layer is computed by stacking the activation maps for all the filters along the depth dimension. Each entry in the output volume is outputted as a neuron's input. The Input Layer serves as a buffer for the input before it is sent to the following layer. Here in the buffer, we read the image and the main operation of feature extraction is performed by the Convolution Layer. It uses the input data to conduct a Convolution process. By starting the kernel over the input, the convolution procedure achieves some of the product at each place. ReLU is an activation function that is used to inject nonlinearity into a system. It returns negative values with zero, allowing the learning process to be stimulated. The activation function receives the output of each convolution layer. The first experiment was training the CNN with all the images and the batch size of 3x3. The number of epochs was 10 and the result for this was 70% percent of recognition. By increasing the number of epochs to 18, the system manages to obtain a 79% percent of recognition. After 18 epochs the system tends not to go more than 79-80% and hits the overfitting. In order to reduce overfitting, we are applying Dropout at the input of some Conv layers (with dropout probability equal to 0.2). In order to normalize the inputs to the nonlinear activation function, batch normalization is used.

- **Convolutional Recurrent Neural Networks (CRNNs) –** an algorithm which involves CNN followed by the RNN to process images containing sequence learning such as letters. Note that RNNs are designed to work seamlessly with sequences of data, such as text, genome, and handwriting, considered capable of capturing temporal information [42]. The model used for recognition letters with this algorithm is formatted by three layers: the convolutional layer which extracts features through the CNN layer; the recurrent layer splits the features into a certain size and inserts them into the input of bidirectional LSTM having 32 hidden inputs (cells); and the last layer is the transcription layer that makes the conversion of feature-specific predictions to labels using connectionist temporal classification. The convolutional layers are used to extract feature sequences from the input images. This output is passed on to the recurrent layers for making predictions for each frame of the feature sequence. Finally, the transcription layer or the Connectionist Temporal Classification (CTC) is used to translate the per-frame predictions by the recurrent layers into a label sequence. Finally, the output is filtered by a Dense layer. Also, we use the Adam optimizer and 18 training epochs. The convolution network is generated by using the standard convolutional layers along with ax-pooling layers from the CNN model. With the help of weights and biases, it selects features from all images (the input). A bidirectional Recurrent Neural Network receives CNN's output (RNN). For each block in the feature sequence, the recurrent layer is used to predict a label. There are three benefits of using the recurrent layer. Firstly, it can extract contextual data from a sequence. When predicting a label for any letter, it may be easier to distinguish it by mixing it with surrounding alphabets rather than evaluating each alphabet separately. Secondly, the error can be propagated back to the convolutional layers, allowing the entire CRNN model to be trained using a single loss function. Finally, the RNN layer can be fed input of any length. RNN produces a matrix containing scores for each individual character at each time step. As input, CTC receives this matrix together with the appropriate ground-truth text (transcript of the word in the image). It then sums all of the scores generated by using all potential alignments of the ground-truth text in the image. As a result, the score of that ground-truth text is determined. The plan for the baseline was to train a CRNN model which will have a recognition rate under the CNN one. But by surprise, using the same number of epochs over the same data input of the 3x3 input frame, the percentage was a little bit over 80%. The roles are switched in favor of CRNN.

4. Results

The features presented above were subsequently trained with combinations of neural networks. Each model is evaluated from the same data set. It was tried to normalize the image classes by the smallest class having a total of 26000 images. Because of the lack of data, the results were disappointing.

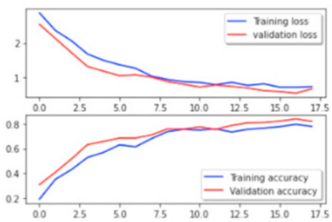

Fig. 5. Loss and accuracy curves for training and validation CNN.

Alternatively, an unnormalized dataset in a total of 372451 images was attempted. Applying this model to this dataset we can observe in the next plot (Fig. 5) the loss and accuracy curves for training and validation that shows how this model performed.

The curves for validation and training lost, in this case, tend to increase for accuracy and decrease for loss. In the next plot, the loss and accuracy curves for training and validation describe how this model performed (Fig. 6).

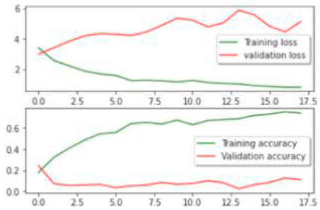

Fig. 6. Loss and accuracy curves for training and validation CRNN.

The tendency of the loss is to go down in both models. Loss value implies how well or poorly a certain model behaves after each iteration of optimization. Ideally, after each iteration, we would expect the loss to be reduced.

To validate this OCR system, Table 2 represents the recognition percentage interval of both models having all images in 26 classes and each class having an unequal number of images and the final decision regarding the system decision. Having the CNN as a baseline and comparator of CRNN, we can say that both scores have almost the same percentage, but with a bit high accuracy for the CRNN system. This can be explained because of the composition of CRNN if CNN and RNN should do an increase of accuracy. The learning phase was formed under 18 epochs.

Table 2. Results for both models – 18 epochs and 372451 images.

Model	Accuracy
Baseline: CNN	80%
CRNN	82%

Table 3 represents the recognition percentage interval of both systems having all images in 26 classes and each class having an equal number of images of 1000 per class. The learning phase was formed under 18 epochs. This shows how important it is to have a large and diversified corpus that can help out not to reach the overfitting step too soon and have better accuracy.

Table 3. Results for both models – 18 epochs and 26000 images.

Model	Accuracy
Baseline: CNN	72%
CRNN	75%

Table 4 represents the recognition percentage interval of both systems having all images in 26 classes and each class having an unequal number of images. The learning phase was formed under 10 epochs. This was the first run for each program. Watching the plot from fig. 6 and 7, it was noticeable that there is room for increasing the accuracy by increasing the number of epochs without having high accuracy, and it was clear that we can add more epochs to increase the accuracy.

Table 4. Results for both models – 10 epochs and 372451 images.

Model	Accuracy
Baseline: CNN	70%

CRNN	73%

Regardless of the number of entries, both models behave appropriately in terms of accuracy, due to the fact that we have the same input options like the number of epochs, the optimizer, and the batch size. That seems to have a huge impact on the system and influence the experimental results. Another major impact that reveals the accuracy is the dataset, which was diversified enough and added a bonus on helping the system not to reach the overfit step.

5. Discussion

Even if it is only a pilot study, we focused on the distinctive nature of handwriting Romanian character recognition in the medical field with the implication of NLP techniques effective enough to this task. Few know that Romanian language has five special letters (Fig. 7), called diacritics (two of them are associated with the same sound), formed by modifying other Latin letters.

Ăă Ââ Îî Șș Țț

Fig. 7. Five Romanian diacritics

Note that most of the time we get 2 out of those 3 wrongs: Ș and Ț because of some glyph substitutions: letter A with tilde or letter A with caron instead of letter A with breve; letter S with cedilla instead of letter S with comma below, and letter T with cedilla instead of letter T with comma below. Of course, ignoring the diacritics, a new problem arises, that of word sense disambiguation. Therefore, at this time our system fails to completely solve the problem of recognizing all handwritten characters by the Romanian physician.

We are developing a set of annotation guidelines for diacritics in order to be used by five computational linguistics domain specialists. At this point, we did the training just for all letters from A-Z, in separate folders, which are contained in the Kaggle library.

6. Conclusions

Due the medical-records scanning can be costly and time-consuming for healthcare organizations that want to stop using paper, AI proves to be the ideal solution for this process. In fact, this study offers significant information about how AI is enhancing medical records scanning. What about the fact that the doctors have often an unreadable handwriting?

The objective of this research has been to implement an OCR application for offline character recognition (a pilot system) that can be used in the medical sector with the implication of NLP techniques. Towards this goal, this paper contributes in two main steps: (a) develop a Romanian medical term corpus and (b) use a deep learning approach for final recognition.

Our approach was designed for recognizing particularly doctors' cursive handwriting and converting them into digital printed texts. A combination CNN + RNN was used to create an offline character recognition system for predicting doctors' handwriting. This research achieved around 80.0% average accuracy using CNN and CRNN. The current accuracy needs to be improved, by considering all Romanian letters, including diacritics and increasing the dataset.

Moreover, these results provide an interpretation which can be useful for other similar studies centered on one of the components of Google Vision architecture, the Bi-LSTM part.

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